

Can Life Keep Up When the Planet Changes Too Fast? Scientists Have a New Answer

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Imagine you're in a race, but the finish line keeps moving faster and faster. That's what it's like for animals when the environment changes too quickly. Scientists at MIT and the University of Leicester wondered: can this idea explain why huge numbers of species sometimes go extinct all at once? They built a math model to test it. Think of a **bell-shaped curve**: most animals **adapt** at a medium speed, but only a few can **adapt** super fast or super slow. When the environment shifts too quickly, the **slowpokes** can't keep up, and they disappear. The team checked their model against 27 big environmental changes over the last 450 million years. They looked at how fast the **carbon cycle** changed a clue about the climate and compared it to how many animal groups died out. The model worked! It correctly predicted the **severity** of almost every **mass extinction**. So, the next time you hear about climate change, remember: it's not just about warmer weather. It's about whether life can sprint fast enough to survive.

Word Treasure

mass extinction -- An event where a huge number of different species die out all over the planet in a short time.

carbon cycle -- The way carbon moves through the air, oceans, and living things, which affects the climate.

bell-shaped curve -- A graph shaped like a bell, showing most things in the middle and fewer at the extremes.

adapt -- To change slowly over many generations to fit better in the environment.

severity -- How bad or serious something is.

slowpokes -- Animals that adapt very slowly compared to others.

Background you need

Earth's history includes several mass extinctions, such as the one that wiped out the dinosaurs 66 million years ago.

Researchers from MIT (a leading U.S. science university) and the University of Leicester investigated whether a too-fast changing environment is to blame.

They looked at the carbon cycle -- how carbon moves between the air, oceans, and living things -- as a clue to past climate speed.

Break it down

WHO: A team of scientists from MIT and the University of Leicester

WHAT: A mathematical model showing that when the environment changes too quickly, slow-adapting species go extinct, leading to mass extinctions

WHERE: Studied data from 27 past events across the globe, covering 450 million years

WHY: To explain past extinctions and warn that today's rapid climate change could be just as dangerous

Why it matters

Understanding how fast change can cause extinctions helps you see why slowing down climate change is important to protect animals we love, like polar bears and butterflies, and even our own future.

Test what you remember

1. What did the scientists from MIT and the University of Leicester study?

- ☐ (A) Why animals grow faster
- ☐ (B) How mass extinctions happen when the environment changes too quickly
- ☐ (C) How dinosaurs built nests
- ☐ (D) Why the ocean is salty

2. What shape did the scientists use to think about adaptation speeds?

- ☐ (A) A circle
- ☐ (B) A square
- ☐ (C) A bell-shaped curve
- ☐ (D) A triangle

3. What happens to the slowpokes when the environment shifts too fast?

- ☐ (A) They get faster
- ☐ (B) They can't keep up and disappear
- ☐ (C) They learn to fly
- ☐ (D) They change color

4. How many big environmental changes did the scientists check?

- ☐ (A) 10
- ☐ (B) 27
- ☐ (C) 50
- ☐ (D) 100

5. What clue did they use to understand the climate's speed?

- ☐ (A) The moon's phases
- ☐ (B) The carbon cycle
- ☐ (C) Volcanic eruptions
- ☐ (D) Cloud patterns

6. What did the math model correctly predict?

- ☐ (A) The number of volcanoes
- ☐ (B) The severity of mass extinctions
- ☐ (C) The time of the next ice age
- ☐ (D) The size of dinosaurs

Answer key (try the quiz first!): 1: B 2: C 3: B 4: B 5: B 6: B

Questions to think about

- 1. Hopeful:** The model could help scientists identify which animals need the most help to survive a warming planet.
- 2. Critical:** Some might argue that if extinctions are natural, we don't need to change our behavior, but scientists say today's changes are human-caused and unusually fast.
- 3. Scientific:** The bell-curve idea shows that most species adapt at a medium pace, so when the climate outpaces that, many can't keep up -- a simple but powerful warning.

MY THOUGHTS (at least 20 words):